Failure Analysis Report

FAIL-0605 — Timeout

1. Failure Summary

Failure ID	FAIL-0605
Failure Type	Timeout
Severity	Minor
Component	Database
Application	APP-005 — WearableHealth
Build Version	5.5.9
Timestamp	2023-01-31 23:49:24
Environment	ENV-002 — Wearable Lab
OS Version	Wear OS 4
Network	Bluetooth 5.3
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-1209
Title	Ensure that AR scene rendering does not exceed the memory ceiling (256MB) thresh...
Execution ID	EXEC-03085
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0412 — Security — Offline Mode

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete media streaming within the defined SLA.
Observed Behaviour	The system exhibited timeout when the battery level drops below 10%.

4. Error Details

Error Message:

```
Operation timed out after 5570ms; threshold is 8641ms.
```

Stack Trace / Log Excerpt:

```
com.samsung.smarthome.PaymentGateway.execute(line 583)
```

Log Reference	LOG-90837

5. Executed Steps at Time of Failure

1	Navigate to Settings > Account and tap 'Verify Identity'.	The audit log records the event with correct metadata.

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2	Verify that the UI displays 'Sync complete'.	The performance counter logs a value below the SLA threshold.
3	Verify that the UI displays 'Payment approved'.	A confirmation notification appears on the paired device.
4	Launch the WearableHealth application on the test device.	Server returns HTTP 200 with a valid JSON payload.

6. Root Cause Analysis

Retry logic does not apply exponential back-off; storm of reconnection attempts saturates the gateway.

8. Historical Similar Failures

HIST-0604	FAIL-0605	Timeout	3.5.1	2022-09-05	5	Resolved
HIST-0803	FAIL-0804	Timeout	6.5.0	2022-03-24	1	Open
HIST-0950	FAIL-0951	Timeout	3.1.6	2023-05-19	11	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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